

AP CSP | Lesson 02: Computing Innovations

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C is correct: $16 + 8 + 2 = 26$. A omits the 2 position; B treats the final 10 as 1 instead of 2; D replaces the 2 contribution with 4. Leading zeros do not change the value.

Q2. Answer: B

B is correct: $2^5 = 32 < 35 \leq 64 = 2^6$. A has too few patterns. C works but is not minimal. D confuses the number of rooms with the number of bits needed to encode each room.

Q3. Answer: D

D identifies a user goal that the innovation can serve. A describes inputs, B describes an output, and C describes processing. These three explain aspects of function rather than the underlying user need.

Q4. Answer: C

C transforms sound through software and can serve creative expression. The stated A operates mechanically; B and D do not execute a program. A device need not be connected to the internet to be a computing innovation.

Q5. Answer: A and C

A and C are data supplied for processing. B is an output from this program. D is part of its program logic, not the supplied moisture data. Award one mark only for A and C together, with no other choices.

Q6. Answer: B

B is correct: before the later click the stored count can be 8. A is false because a click event can supply input. C adds an unsupported cause; the stated program rule is sufficient. D overlooks prior state and would contradict the stated behavior.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) Help readers find books on subjects they want to explore. (b) Use the selected topic to search the catalog and display matching titles. (c) Input: the selected topic (or catalog data read by the app). Output: the displayed matching titles.

1 mark: States a plausible purpose that identifies readers and their information or reading goal.

1 mark: Describes using the topic to find matching catalog entries and display titles.

1 mark: Identifies a valid input data item: selected topic or catalog data read by the app.

1 mark: Identifies the matching titles as output.

Q8. 4 marks

Answer: (a) The temperatures 18, 21 and 24, read from a file. (b) Add them and divide by three. (c) The text "Mean: 21" written to the results file. (d) The program sends result data to a storage device; output is not restricted to a screen.

1 mark: Identifies all three temperatures as the input and the file as their source.

1 mark: Describes finding the mean by adding the three values and dividing by three.

1 mark: Identifies "Mean: 21" or the mean value 21 written to the results file as output.

1 mark: Explains that writing result data to storage is output and does not require screen display.

Q9. 4 marks

Answer: (a) “Occupied” is output from the sensor program and input to the room finder. (b) The room finder excludes the occupied room from its list. (c) It displays the revised list of available rooms.

1 mark: Classifies the message as output from the sensor program.

1 mark: Classifies the same message as input to the room finder.

1 mark: Identifies removal or exclusion of the occupied room as processing.

1 mark: Identifies the revised available-room list as the displayed output.

Q10. 4 marks

Answer: (a) The “Add one” button-click event. (b) It displays 3 or 9, respectively. (c) Output depends on the prior stored tally as well as the click. (d) The new tally is data: a value processed and stored by the program.

1 mark: Identifies the click event on “Add one” as the input.

1 mark: Gives both displayed results correctly: 3 and 9.

1 mark: Explains that different prior stored tallies cause the outputs to differ.

1 mark: Classifies the new tally as data, a value the program handles, rather than an instruction.

Q11. 4 marks

Answer: (a) “What made it difficult to find a suitable study room last time?” (b) Input: requested group size. Output: names of available rooms with enough seats. (c) Example: for a group of 4, when the only available rooms have capacities 2 and 6, display only the room with 6 seats.

1 mark: Gives an interview question capable of revealing a user need, experience or constraint.

1 mark: Names a relevant input data item, such as requested group size.

1 mark: Names a useful output data item, such as a list of suitable available rooms.

1 mark: States a requirement specific enough to check, including conditions and an observable expected result.

Q12. 4 marks

Answer: (a) For 2: A and B. For 7: no matching room. (b) C has enough seats but is unavailable, so displaying it ignores the availability requirement. (c) Display “No suitable room available” or an equivalent clear message.

1 mark: Gives both A and B, and no C, for a group of 2.

1 mark: Gives no matching room for a group of 7.

1 mark: Explains that C must be excluded because it is unavailable despite its capacity.

1 mark: Proposes a clear no-match output message consistent with the requirement.